

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Satyala Murali Karthik
Roll Number	MCL2026010
Programme / Department	Cyber laws and information security
Topic ID Full Assigned Topic	T21 Benchmarking Citation Fabrication Rates Across Large Language Models and Agentic Search Tools
AI Tool Used	Google Gemini
Date	21/08/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified? Question 2 -

Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
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A Record your baseline perception of AI citations AI-Assisted Citation Integrity Audit | Parts A-G

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B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F G	Judge their plausibility before verification Verify authenticity, metadata and identifiers

H Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☒ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☒ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☒ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Google Gemini
Model name	Flash
Model/version shown Access tier	3.6 ☒ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available? Was web browsing actually used? Research/Deep Research mode used?	☒ Yes ☐ No ☐ Not sure ☒ Yes ☐ No ☐ Not sure ☐ Yes ☒ No
Date/time generated	21/08/2026

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Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

☒ Save the complete AI-generated paper

☒ Save the complete reference list

- ☒ Take screenshot(s) showing the generated output/references
- ☒ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Benchmarking Citation Fabrication Rates Across Large Language Models and Agentic Search Tools

Measure	Value
Approximate pages	7
Approximate word count	≈1700
Number of sections	10
Total number of references	3
Approximate in-text citation occurrences	10
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☒ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	1
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	2

URL only 0
No persistent identifier supplied 0

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TOTAL	3

PART E - SELECT 10 REFERENCES FOR AUDIT

Audit Slot	Original Bibliography Reference No.
R1	1 — Li, M., Lin, Z., & Ma, S. (2026)
R2	2 — Walters, W. H., & Wilder, E. I. (2023)
R3	3 — Zhao, Z., & Yin, Y. (2026)

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Specific-sounding title, named authors, arXiv preprint format	4	Yes
R2	Real journal ,full DOI, specific volume/article number	5	Yes
R3	Plausible topic, arXiv preprint format but shares the identical URL with R, which is suspicious on its face	3	No

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental

observations. AI-Assisted Citation Integrity Audit | Parts A-G

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

- ☒ Persistent identifier: DOI / arXiv / PMID
- ☒ Publisher or official repository
- ☒ Google Scholar
- ☒ Semantic Scholar / OpenAlex
- ☒ Crossref
- ☒ PubMed or another discipline-specific database
- ☒ Exact-title search
- ☒ Author + title-keyword search

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Yes	Yes	Yes	Yes	Yes (arXiv)	No
R2	Yes	Yes	Yes	Yes	Yes	Yes
R3	Yes	Yes	Partial (2/6 authors matched)	Yes	Yes	Yes

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	arXiv direct search, author homepage	arXiv:2605.07723	Title/authors ("Mingzhe Li, Zhiqiang Lin, Shiqing Ma," UMass Amherst/OSU) belong to a real paper — but that paper's actual arXiv ID is 2605.08583, confirmed

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			via arXiv abstract page and the first author's homepage. The ID cited (2605.07723) resolves to an entirely different paper (see R3).
R2	PubMed, Nature/Scientific Reports, Semantic Scholar, EBSCO	DOI: 10.1038/s41598-023-41032- 5	Exact match on all fields: title, authors, journal, volume 13(1), article 14045, 2023. DOI resolves correctly to the paper on nature.com.

ID resolves correctly to "LLM hallucinations in the wild: Large-scale evidence from non-existent citations," submitted 8 May 2026. Title matches exactly. Real

R3arXiv direct search,
Phys.org, ResearchGatearXiv:2605.07723 (as cited)
authors are Zhenyue Zhao, Yihe Wang, Toby Stuart, Mathijs De Vaan, Paul Ginsparg, Yian Yin (6 authors) — the citation lists only "Zhao, Z., & Yin, Y.," omitting 4 of 6 authors without an "et al."

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	E	Real paper by Li, Lin & Ma exists, but the cited arXiv ID (2605.07723) resolves to a different paper by Zhao et al., correct ID is 2605.08583.

R2	A	All fields — title, authors, year, journal, DOI — verified exactly against the publisher record.
R3	B	Paper exists and ID/title/year are correct, but the author list is truncated to 2 of the actual 6 authors, misrepresenting authorship.

PART G - SCORING

Count your final classifications: A = 1 B = 1 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 3

Code	Points
A	4
B	3
C	1

D 0
E 1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

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My calculation: $4(1) + 3(1) + 1(0) + 0(0) + 1(1) = 8$ points

Authenticity Score = $8 / (4 \times 3) \times 100 = 66.67 / 100$

Score = 60-74 Moderate citation-authenticity risk

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = $(\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$

Correct predictions = 2 Total predictions = 3 Prediction Accuracy = 66.67 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	3
References deeply audited	3
A: Verified	1
B: Wrong metadata	1
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	1
Authenticity Score	66.67 /100
Prediction Accuracy	66.67%

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

Partially. All three references pointed to real papers by real authors there was no outright fabrication or Frankenstein blending here. But "genuine underlying paper" didn't mean "accurate citation": one reference's identifier silently pointed to the wrong paper, and another dropped 4 of 6 authors. Professional appearance masked structural errors rather than covering total invention. .

2. Did all genuine references actually support the claims for which they were cited?

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Not fully assessed here but notably, the content actually matches well: Walters & Wilder (2023) genuinely does report the ChatGPT-3.5/4 fabrication rates the paper describes, and the real Zhao et al. paper genuinely does describe large-scale citation-hallucination audits so the underlying claims are reasonably well-supported by the correct papers, even where the citation mechanics (IDs, author lists) were flawed.

3. What was the most serious citation-integrity problem you observed?

The identifier collision: both "Li, Lin & Ma" and "Zhao & Yin" were cited with the identical arXiv URL (2605.07723). That ID genuinely belongs to Zhao et al.'s paper meaning the Li/Lin/Ma citation is an Identifier Hijacking case (real paper, wrong ID pointing at an unrelated real paper), which is arguably more dangerous than outright fabrication since a careless reader who clicks the link finds a real, credible-looking paper and may never notice it's the wrong one.

4. What is the single most important lesson you learned from this lab?

A reference "existing" is not the same as a reference being correctly cited. The most convincing-looking citations in this sample weren't the safest the duplicate-URL problem was only catchable by cross-checking references against each other, not by researching any single reference in isolation. Auditing citations one at a time can miss systemic errors that only show up in relational patterns across the full bibliography

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

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Roll Number: MCL2026010

S. M. Karthik

Signature:

Date: 21/8/2026

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